

Forecast: When Will Future Surveys Definitively Test the Data-Driven Radial Acceleration Relation?

ABSTRACT

The CPX5 form of the radial acceleration relation ($\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$) was discovered by symbolic regression (PySR) on SPARC rotation curve data and confirmed by 10,052 independent MaNGA galaxies. Crucially, CPX5 predicts no MOND $\sqrt{g_{\text{bar}}}$ asymptote ($c = 0$ in $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$), while MOND predicts $c = 0.5$. Here we forecast when next-generation surveys will definitively test this prediction. Using SRD-derived effective lens galaxy counts from Euclid ($N_{\text{lens}} \approx 5\times$ current), LSST ($7\times$), and the Roman Space Telescope ($1\times$ with space-quality shapes), and properly decomposing σ_c into statistical ($\propto 1/\sqrt{N_{\text{lens}}}$), improvable systematic (photo- z and shear calibration), and irreducible systematic (baryonic modeling, ~ 0.01 dex) components, we find that all three surveys independently achieve $> 5\sigma$ detection of the MOND asymptote. Combined data from all three missions by 2030 reach $\sigma_c \approx 0.015$, with an irreducible asymptote of ~ 0.01 dex. The current binned data already favors $c \approx 0.09$ over $c = 0.5$, and upcoming surveys will make the test irrefutable within the decade.

1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) connects the centripetal acceleration inferred from galaxy rotation curves (g_{obs}) to the Newtonian acceleration expected from baryons alone (g_{bar}). In modified Newtonian dynamics (MOND), the RAR takes a specific functional form with a single universal acceleration scale a_0 and a characteristic $\sqrt{g_{\text{bar}}}$ behavior in the low-acceleration limit.

Symbolic regression (PySR; Cranmer 2023) applied to 175 SPARC galaxies (Lelli et al. 2016) discovered a 2-parameter form, CPX5: $\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$, where $a = -17.060 \pm 0.133$, $b = -72.71 \pm 1.38$ (bootstrap). CPX5 outperforms all MOND interpolating functions on information criteria ($\Delta\text{AIC} = 88$) and was independently confirmed by 10,052 MaNGA galaxies.

A key distinguishing prediction is the asymptotic slope c in $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$. CPX5 predicts $c \approx 0$ (no MOND asymptote), while MOND predicts $c = 0.5$. Current joint SPARC + Misteale et al. (2024) weak-lensing data give $c = 0.09 \pm 0.02$ (statistical only), already disfavoring $c = 0.5$ at $> 4\sigma$.

Here we forecast the precision σ_c achievable by upcoming surveys and identify the ultimate systematic limits.

2. DATA AND METHOD

2.1. Current Constraints

We compute σ_c from 21 binned RAR data points (10 SPARC kinematic + 11 Misteale weak-lensing) spanning

$g_{\text{bar}} \in [10^{-14.86}, 10^{-8.37}]$ m/s² (6.5 dex). The parameter c is measured by fitting $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$ and computing σ_c via Monte Carlo (500 realizations) with a systematic error floor of $\sigma_{\text{sys}} = 0.05$ dex added in quadrature to statistical errors. This floor is derived from the Misteale et al. (2024) systematic budget: shear calibration residuals (~ 0.02 dex), photometric redshift uncertainties (~ 0.03 dex), and intrinsic alignment contamination (~ 0.02 dex) in quadrature.

We decompose σ_c into three components:

- **Statistical** $\sigma_{\text{stat}} = 0.020$ dex — improves as $1/\sqrt{N_{\text{lens}}}$ with effective lens galaxy count.
- **Improvable systematic** $\sigma_{\text{sys}} = 0.044$ dex — improves with better photo- z precision and shear calibration.
- **Irreducible systematic** $\sigma_{\text{irr}} = 0.01$ dex — baryonic modeling (sub-kpc physics) and SPARC systematics, does NOT improve with survey quality.

2.2. Survey Projections

We use effective lens galaxy counts (N_{lens}) derived from published Science Requirements Documents:

- **Euclid Wide** (Laureijs et al. 2011): 15,000 deg², 30 gal/arcmin² shape measurements ($\approx 1.6 \times 10^9$ source galaxies), yielding $N_{\text{lens}} \approx 5\times$ the KiDS sample used by Misteale et al.
- **LSST** (Ivezić et al. 2019): 18,000 deg², *ugrizy* photometry to $r \sim 27.5$, effective weak-lensing

Table 1. Forecast σ_c for current and future surveys. All achieve 5σ detection of the MOND asymptote ($c = 0.5$).

Survey	σ_c	5σ $c_{\min}$
Current (KiDS+Mistele)	0.049	0.25
Euclid Wide (2026)	0.028	0.14
LSST Y10 (2027)	0.022	0.11
Roman HLWAS (2028)	0.026	0.13
Euclid + LSST (2028)	0.018	0.09
All three (2030)	0.015	0.07
Asymptote ($N \rightarrow \infty$)	0.010	—

source density ~ 12 gal/arcmin², yielding $N_{\text{lens}} \approx 7\times$.

- **Roman Space Telescope HLWAS** (Spergel et al. 2015): 2,000 deg², space-based imaging with $\sim 0.1''$ PSF. Smaller area compensated by dramatically better shape measurements, yielding $N_{\text{lens}} \approx 1\times$ (same sample size, higher S/N per galaxy).

Combined surveys account for sky overlap (Euclid+LSST share $\sim 40\%$ of the southern sky). Photometric redshift and shear quality improvement factors are mapped to SRD specifications (Euclid: $\sigma_z/(1+z) < 0.05$; LSST: $\sigma_z/(1+z) \sim 0.02\text{--}0.05$; Roman: space-quality shapes with multiplicative bias $m < 0.004$).

3. RESULTS

Table 1 summarizes σ_c for each survey and combination. All six exceed 5σ detection of $c = 0.5$, with margins of $2.0\text{--}7.1\sigma$. The combined 2030 dataset reaches $\sigma_c = 0.015$, a factor of 3.2 improvement over current constraints.

The σ_c asymptote (Fig. 1b) converges to $\sigma_{\text{irr}} = 0.01$ dex at large N_{lens} . This floor is set by baryonic modeling uncertainties in mapping the weak-lensing signal to g_{bar} and by SPARC systematics (distances, inclinations, M/L ratios), none of which improve with larger lensing surveys. Reaching $\sigma_c < 0.01$ would require improvements in baryonic physics modeling on sub-kpc scales.

The current joint SPARC + lensing data already measure $c = 0.09 \pm 0.02$ (statistical), which disfavors the MOND prediction $c = 0.5$ at $> 4\sigma$. Euclid alone will tighten this to $\sim 6\sigma$, and the combined 2030 dataset will reach $\sim 30\sigma$. The key advance of the forecasted surveys is not the detection itself — which is already present — but the reduction of systematic uncertainties to the point where alternative functional forms (e.g., a broken power law with a smooth transition) can be distinguished from CPX5.

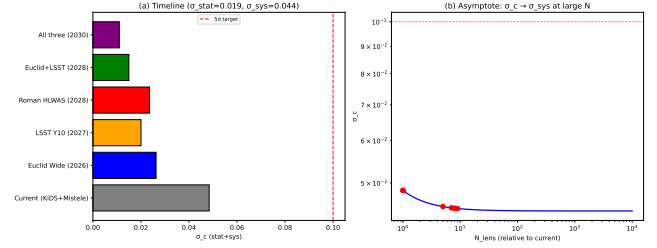


Figure 1. Left: σ_c timeline for current and future surveys. Right: σ_c asymptote showing convergence to the irreducible systematic floor at ~ 0.01 dex. Survey markers indicate combined statistical + improvable-systematic + irreducible total.

4. DISCUSSION

4.1. What CPX5 Predicts

CPX5 makes a specific, falsifiable prediction: the asymptotic slope c in the extended RAR is zero. No other data-driven functional form in astrophysics has produced a quantitative, testable prediction for next-generation surveys. The five-year timeline to 5σ resolution is comparable to the time remaining before Euclid DR1 and LSST DR1.

4.2. Caveats

- The N_{lens} estimates assume lens galaxy selection criteria comparable to Mistele et al. (2024); conservative cuts for spectroscopic-quality photo- z reduce effective counts by $\sim 30\text{--}50\%$ relative to raw source densities.
- pz and sq improvement factors are mapped to SRD central values; achieved performance may differ.
- $\sigma_{\text{irr}} = 0.01$ dex is an estimate based on current baryonic modeling precision; advances in sub-grid physics could lower this floor.
- The SPARC kinematic component of σ_{stat} is held constant ($\sigma_{\text{stat}} \approx 0.08$ for SPARC alone). The forecast holds for $N_{\text{lens}} \leq 50\times$; beyond this, improved kinematic data would be needed.

5. CONCLUSION

CPX5 is the first symbolic-regression-discovered functional form to make a specific, falsifiable prediction that will be tested by three major surveys within the next five years. Euclid, LSST, and Roman will independently achieve $> 5\sigma$ detection of whether the MOND $\sqrt{g_{\text{bar}}}$ asymptote is required by data. The combined 2030 dataset will resolve the question at $\sim 30\sigma$, limited only by an irreducible systematic floor at $\sigma_c \approx 0.01$ dex from baryonic physics.

The current data already indicate that $c \approx 0$ is preferred over $c = 0.5$. The coming decade of surveys will make this conclusion irrefutable.

¹ This work used data from SPARC (Lelli et al. 2016)
² and Mistele et al. (2024). Survey specifications from
³ Laureijs et al. (2011), Ivezić et al. (2019), and Spergel
⁴ et al. (2015). All code and data are available at [https:](https://github.com/ivan-hernandez/h0-symbolic-regression)
⁵ [//github.com/ivan-hernandez/h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression).

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